

- 5 (a) at $t = 1.0 \text{ s}$, $V = 2.5 \text{ V}$ C1
 energy = $\frac{1}{2}CV^2$ C1
 $0.13 = \frac{1}{2} \times C \times (8.0^2 - 2.5^2)$ M1
 $C = 4500 \mu\text{F}$ A0 [3]
- (b) use of two capacitors in series in all branches of combination M1
 connected into correct parallel arrangement A1 [2]

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6 (a) parallel (to the field)

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